1.

(a) $\nabla f(x) = Ax + b$

(b) $\cancel{g}(\nabla h(x)) = g'(h(x))$? I guess

Consider $\frac{\partial g}{\partial x_i} = \frac{\partial g}{\partial h} \cdot \frac{\partial h}{\partial x_i}$

So $\nabla \cdots\cdots = g'(h(x)) \cdot \nabla h(x)$

(c) $\nabla^2 f(x) = A$

(d)

$$\nabla f(x) = \begin{bmatrix} \frac{\partial}{\partial x_i} g(a^T x) \\ \cdots\cdots \end{bmatrix}$$

$$\frac{\partial}{\partial x_i} g(a^T x) = \frac{\partial}{\partial x_i} g\left(\sum a_k x_k\right)$$

$$= g'(a^T x) \cdot a_i$$

$$\nabla f(x) = g'(a^T x) \cdot a$$

$$\frac{\partial}{\partial x_i} [\nabla f(x)]_j = \frac{\partial}{\partial x_i} \underline{g'(a^T x)} \cdot a_j$$

$$= a_j \cdot g''(a^T x) \cdot a_i + \cancel{g'(a^T x) \cdot 0}$$

$$\boxed{\therefore \nabla^2 f(x) = g''(a^T x) a a^T}$$

2.

(a) $x^T A x = x^T z z^T x$

$= (x^T z)^2 \geq 0$

(b) $N(A) =$ (those aren't λz)

Rank $= 1$

(c) uhh……

$x^T B A B^T x = \underline{(B^T x)^T A (B^T x)} \geq 0$ since A PSD

So it's Right

3.

(a) $A = T \Lambda T^{-1}$ (invertible T)

$\Leftrightarrow AT = T\Lambda$

$A[t^{(1)} \cdots t^{(n)}] = [t^{(1)}\lambda_1 \cdots\cdots t^{(n)}\lambda_n]$

for each column $\rightarrow At^{(i)} = \lambda_i t^{(i)}$

(b) isn't it the same as (a)?

(c) $AU = U\Lambda$

$A[u^{(1)} \cdots u^{(n)}] = [u^{(1)}\lambda_1 \cdots\cdots]$

~~For all~~ $\boxed{x^T A x \geq 0}$

Let $Ut = x$

$(Ut)^T A U t$

$= t^T \underbrace{U^T A U}_{} t$

$= t^T \Lambda t$

$= \sum \lambda_i t_i^2 \geq 0$

只是必要条件……

Oh!

if $x = u^{(i)}$ (that means only a "1" in t)

$t^T \Lambda t$

$= \lambda_i t_i^2 \geq 0 \Rightarrow \lambda_i \geq 0$ apply to all i

~~so~~ $\Lambda_{ii} \geq 0$ for each i